

Questions of Consciousness

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In the summer of 2018, a young orca named Tahlequah¹ gave birth in the waters off the Pacific Northwest. When her calf died within hours, she did something that captivated millions around the world. Using her rostrum and flippers, she carried her baby's body through the ocean for seventeen days and over a thousand miles, refusing to let go. She balanced the small corpse on her head as she swam, dove to retrieve it when it slipped away, and pushed it gently through the waves while her pod waited, watched, and accompanied her in what could credibly be called a funeral procession.

The images spread rapidly across social media and news outlets. People everywhere followed her story, many moved to tears by something they recognized immediately and viscerally. They saw a mother's love, a parent's refusal to accept loss — an expression of mourning that felt intimately familiar despite emanating from a being whose world we can scarcely imagine.

Millions of people perceived a depth of felt experience in Tahlequah, a being radically unlike themselves. That perception was not naive. It is among the most consequential of our intuitions.

But the scientific framework that has dominated our modern view of the world for well over a century has trained us to distrust the recognition of consciousness in other animals. Attributing grief to a non-human animal is dismissed as anthropomorphism — a sentimental projection rather than a legitimate inference. The scientific view holds that complex emotions and cognition are uniquely human characteristics, that human consciousness is supreme — the only form capable of understanding the universe that produced it.

Yet that view carries its own undisclosed premise: the universe is, at bottom, only what physical methods can detect, a position philosophers refer to as physicalism. In the scientific framework, this is far more than a philosophical commitment. It is the neutral starting point of serious inquiry. Alternatives are expected to justify themselves in terms that science accepts — evidence that is measurable and replicable — or be dismissed as unscientific and therefore not worthy of serious consideration. The scientific view presumes that reality can be fully and only understood through measurement.

That presumption has worked extraordinarily well in many ways. Our industrial civilization is largely defined by the discoveries of science and the technologies they have spawned. But the basic premise that everything real can be measured falters when confronting consciousness. And consciousness is unquestionably real. At bottom, it

¹ Tahlequah was the name given by The Whale Museum in Friday Harbor, Washington through their Adopt-a-Whale program. Marine biologists refer to her as "J35".

is the only thing we can be certain of. Tahlequah's journey revealed the gap between this fundamental certainty and a conception of reality limited to what scientific methods can answer. Science can tell us how things behave under controlled conditions. But it cannot tell us anything about what it is like to be a conscious being.

Science can, however, shed light on how widespread consciousness is. And what we find, as we look out across the tree of life, is that consciousness appears to be a common and recurring outcome of evolution. Consciousness is not a single thing. It ranges from the minimal sentience of simple animals to the rich inner lives of social mammals and birds, and the gradient between is continuous rather than step-wise. Forms of consciousness sophisticated enough to support memory, learning, and flexible response have evolved many times, in lineages separated by hundreds of millions of years and built on radically different neural architectures. Toothed whales — Tahlequah's lineage among them — developed their large brains long before our ancestors walked upright. Humans are the latest form of highly sentient life on earth, not its apex. These are more than philosophical claims. They are verifiable observations, and they sit uncomfortably within a framework that treats complex consciousness as a recent development unique to *Homo sapiens*.

I argue that those observations are not anomalies to be explained away but evidence that consciousness is widespread in both type and extent. Whether Tahlequah felt grief is not the question this essay turns on. Why evolution has so frequently produced the capacity for it — many species with independent lineages, across the long arc of evolutionary time — is. And that question opens onto a larger one. What kind of universe so broadly produces minds?

Complexities of Consciousness

Tahlequah is an orca, one of approximately 90 cetacean species — the whales, dolphins, and porpoises — almost all of the mammalian species to become fully aquatic and remain entirely bound to water.² Many toothed-whale species possess remarkably large brains: sperm whales have the largest brains on Earth, reaching about eight kilograms — roughly six times human brain mass. Orca brains weigh around five kilograms, and several other odontocete species equal or exceed humans in absolute brain size. More remarkably, large-brained odontocetes were flourishing in the oceans tens of millions of years ago — long before the earliest hominins walked upright.

Why? Why has evolution repeatedly invested in and sustained such metabolically expensive brain tissue across geological timescales? What purpose does all that neural architecture serve?

² The four sirenian species — three manatees and one dugong — are the only other fully aquatic mammals.

The question broadens and sharpens when we look beyond cetaceans. Cephalopods diverged from vertebrates over 500 million years ago and evolved a fundamentally different neural architecture, yet their problem-solving, exploratory behavior, and apparent sentience make it difficult to treat experience as a vertebrate monopoly. Corvids and parrots developed remarkable cognition with distinctly different brain structures. Elephant brains, by far the largest of terrestrial animals, evolved well before and along completely different lines than primates. Non-human great apes have demonstrated self-awareness and metacognitive monitoring.

The pattern — complex cognition, flexible agency, and plausible forms of rich experience arising independently, through different architectures — is not easily explained as coincidence. When evolution converges repeatedly on the same outcome through radically different means, it suggests something about the landscape of biological possibility itself. Eyes evolved independently dozens of times; flight at least four times; warm-bloodedness at least twice. These convergences are understood as evidence that certain biological solutions are deeply favored — that evolution, while not predetermined, is constrained, channeled, and structured by the landscape of viable forms.

Complex consciousness may be another such convergence. A growing body of work makes that possibility harder to dismiss. In evolutionary theory, paleontology, comparative cognition, affective neuroscience, and complexity science, researchers have increasingly come to see mind as part of evolution's broader convergent pattern. Simon Conway Morris has argued that convergent evolution reveals deep attractors in biological possibility-space. Stuart Kauffman's work on self-organization suggests that living systems generate forms of order that natural selection shapes but does not wholly create.³ Meanwhile, work in animal-sentience science and the philosophy of mind — Peter Godfrey-Smith on cephalopods, Eva Jablonka and Simona Ginsburg on the origins of subjective experience, Todd Feinberg and Jon Mallatt on its neural foundations, Jonathan Birch on its philosophical assessment — has made it increasingly difficult to treat consciousness as an exclusively human, or even exclusively mammalian, phenomenon. These researchers do not share a single theory, and none of them argue that consciousness is inevitable. But together they support a more modest and more important claim: complex consciousness is no longer best understood as a biological freak accident.

If this is even approximately right, it shifts the burden of explanation. Consciousness need not be treated as something that somehow arose in an otherwise mindless process, but as an increasingly elaborate expression of what living systems have long been doing — sensing, integrating, responding, and acting within worlds of their own.

³ This broader reassessment of evolutionary causation has become increasingly organized. In 2023, a group of researchers — including Kauffman, Pross, Denis Noble, James Shapiro, Denis Walsh, and Michael Levin — contributed to a collaborative volume arguing that the evolved purposiveness of living systems has itself been a major causal factor in evolutionary history. The volume emerged from a Linnean Society conference and reflects a growing scientific perspective sometimes called the “Third Way of Evolution” — distinct from both intelligent design and the standard Modern Synthesis accounts. This does not establish that evolution is directed toward consciousness, but it does challenge the older picture of organisms as passive products of random variation and external selection (Peter A. Corning et al., *Evolution "On Purpose": Teleonomy in Living Systems* (MIT Press, 2023)). There is intense debate about this in biological circles: see (Kevin Laland et al., “Does Evolutionary Theory Need a Rethink?” *Nature* 514, no. 7521 (October 2014): 161–64, <https://doi.org/10.1038/514161a>)

The cetaceans are not curiosities. They are evidence.

The Questions Ahead

The standard framing puts the burden of explanation on consciousness: given a universe of matter in motion, how does subjective experience arise? This framing assumes that the natural state of the world is unconsciousness, and that consciousness is the anomaly requiring special explanation. It is so deeply embedded in contemporary thought that it feels less like an assumption than like common sense.

But the evolutionary evidence inverts this. If the convergent emergence of complex minds across independent lineages is not coincidence but tendency — if the universe reliably produces consciousness — then the question reverses: not “How does unconscious matter produce consciousness?” but “What kind of universe produces consciousness as a matter of course?”

That question is the foundation of this essay. The story begins by tracing how we arrived at a framework that makes the question seem strange — how a methodology of extraordinary power hardened into metaphysics, and how the living cosmos that prior civilizations recognized gave way to a universe of dead matter in mindless motion. It then follows the recovery of what was displaced, through the founding generation of quantum mechanics and into the contemporary biological challenge to the mechanistic picture from within science itself. From there it turns to the evidence: the evolution of consciousness across the tree of life and what it reveals about the kind of universe we inhabit.

This essay does not arrive in isolation.⁴ Across philosophy, neuroscience, evolutionary biology, and foundations of physics, careful inquiry from very different starting points has been converging for at least two decades on a shared diagnosis: that consciousness may belong nearer the foundations of nature than the standard story allows. The gathering remains loose — there is no school, no single name. But when inquiry from radically different entry points arrives at the same diagnosis, the convergence is itself evidence. This essay joins that conversation.

My argument is not that physicalism is wrong in the way a false hypothesis is wrong. It is that physicalism is an inadequate story — one that accounts well for what it was designed to account for, and poorly for much that falls outside those boundaries. Science plainly works. But a metaphysical commitment that grew up alongside it — the assumption that only what physical methods can detect is fully real — has been confused with science itself. That confusion has made it difficult to discuss the nature of consciousness without being heard

⁴There is no commonly accepted name for this convergence, and the absence is itself diagnostic — the gathering is real but loose. Contributions from very different starting points include (Thomas Nagel, *Mind and Cosmos: Why the Materialist Neo-Darwinian Conception of Nature Is Almost Certainly False* (Oxford, New York: Oxford University Press, 2012)), in philosophy; (Mark Solms, *The Hidden Spring: A Journey to the Source of Consciousness* (W. W. Norton & Company, 2021)), in neuroscience; (Kevin Mitchell, *Free Agents: How Evolution Gave Us Free Will* (Princeton University Press, 2023)), and (Corning et al., *Evolution "On Purpose"*), in evolutionary biology; and the Pauli–Jung tradition in the foundations of physics, recently developed by Harald Atmanspacher and others (*Dual-Aspect Monism and the Deep Structure of Meaning* (n.d.)). The list is illustrative, not exhaustive.

as arguing against Darwin or for divine intervention. This essay does neither. It is derived from what evolutionary biology and neuroscience have actually found — and from the observation that those findings sit uncomfortably within the framework typically invoked to explain them.

- Corning, Peter A., Stuart A. Kauffman, Denis Noble, James A. Shapiro, and Richard I. Vane-Wright. *Evolution "On Purpose": Teleonomy in Living Systems*. MIT Press, 2023.
- *Dual-Aspect Monism and the Deep Structure of Meaning*. n.d.
- Laland, Kevin, Tobias Uller, Marc Feldman, Kim Sterelny, Gerd B. Müller, Armin Moczek, Eva Jablonka, et al. "Does Evolutionary Theory Need a Rethink?" *Nature* 514, no. 7521 (October 2014): 161–64. <https://doi.org/10.1038/514161a>.
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